

Course Project 1

Turbine–Governor Closed-Loop System I

REE 404 — Energy Generation and Control

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Simulation notebook:

https://colab.research.google.com/drive/17zqGKTEQxnP6pUJPJgUh70_figeg-k0j?usp=sharing

1 System Description

The system studied is the single-area turbine–governor speed-control loop shown in Figure 1. A speed-droop governor ($1/R$) and a reheat steam turbine ($\frac{1+F_{HP}T_{RH}s}{1+T_{RH}s}$) act on the frequency-error signal to produce the mechanical power ΔP_m . This power, less an external load disturbance ΔP_e , drives the generator–load swing dynamics $\frac{1}{2Hs+D}$, whose output is the rotor speed deviation $\Delta\omega$, which is fed back to the input.

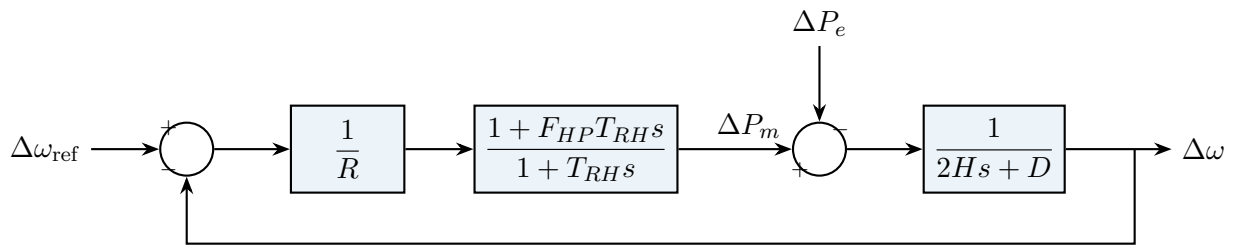


Figure 1. Block diagram of the turbine–governor closed-loop system.

Table 1. Parameters of the turbine–governor system.

Parameter	Value
R (droop)	0.1
T_{RH} (reheat time)	5 s
F_{HP} (HP fraction)	0.3
H (inertia)	3 s
D (damping)	1

Project tasks

1. Derive $\omega(s)/\omega_{\text{ref}}(s)$ step by step.
2. Find the characteristic equation and solve for the poles.
3. Convert the closed-loop transfer function to a minimal state-space representation.
4. Confirm that the eigenvalues of the state-space model match the poles.
5. Frequency-response (Bode) analysis and stability.
6. Step response: overshoot, settling time and steady-state error.
7. Parameter sensitivity: effect of the reheat constant T_{RH} .
8. Python script to derive, compute and simulate, confirming the hand calculations.

2 Task 1 — Closed-Loop Transfer Function

The governor together with the reheat turbine forms the controller

$$G_c(s) = \frac{1}{R} \frac{1 + F_{HP} T_{RH} s}{1 + T_{RH} s} = \frac{1}{0.1} \cdot \frac{1 + (0.3)(5)s}{1 + 5s} = 10 \frac{1.5s + 1}{5s + 1} = \frac{15s + 10}{5s + 1}.$$

The generator–load swing dynamics form the plant

$$G_p(s) = \frac{1}{2Hs + D} = \frac{1}{2(3)s + 1} = \frac{1}{6s + 1}.$$

The open-loop transfer function is

$$G_{ol}(s) = G_c(s) G_p(s) = \frac{15s + 10}{(5s + 1)(6s + 1)} = \frac{15s + 10}{30s^2 + 11s + 1}.$$

With $\Delta P_e = 0$, the closed-loop transfer function is

$$T(s) = \frac{\omega(s)}{\omega_{\text{ref}}(s)} = \frac{G_{ol}(s)}{1 + G_{ol}(s)} = \frac{15s + 10}{(30s^2 + 11s + 1) + (15s + 10)} = \boxed{\frac{15s + 10}{30s^2 + 26s + 11}}.$$

3 Task 2 — Characteristic Equation and Poles

The characteristic equation is the denominator set to zero,

$$30s^2 + 26s + 11 = 0.$$

Solving with the quadratic formula,

$$s = \frac{-26 \pm \sqrt{26^2 - 4(30)(11)}}{2(30)} = \frac{-26 \pm \sqrt{676 - 1320}}{60} = \frac{-26 \pm \sqrt{-644}}{60} = -\frac{13}{30} \pm \frac{\sqrt{161}}{30} j.$$

$$\boxed{s_{1,2} = -0.4333 \pm 0.4230 j}$$

The poles form a **complex-conjugate pair** with a negative real part, so the system is **stable and under-damped**. The natural frequency and damping ratio are

$$\omega_n = \sqrt{\frac{11}{30}} = 0.606 \text{ rad/s}, \quad \zeta = \frac{26/30}{2\omega_n} = 0.716.$$

4 Task 3 — State-Space Representation

Writing $T(s)$ in monic form, $T(s) = \frac{0.5s + 0.3333}{s^2 + 0.8667s + 0.3667}$, the minimal (controllable-canonical) realization is

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{11}{30} & -\frac{26}{30} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} \frac{10}{30} & \frac{15}{30} \end{bmatrix} = [0.3333 \quad 0.5], \quad D = 0.$$

The two states are the internal reheat/speed states, with $x_2 = \dot{x}_1$.

Note (corrected). The output matrix must reproduce the transfer-function numerator: $C(sI - A)^{-1}B = \frac{0.5s + 0.3333}{s^2 + 0.8667s + 0.3667} = T(s)$. Using $C = [15 \quad 10]$ (as in the original draft) does *not* give $T(s)$; the correct row is $C = [10/30 \quad 15/30]$.

5 Task 4 — Eigenvalue Verification

The eigenvalues of A are the roots of $\det(sI - A) = s^2 + \frac{26}{30}s + \frac{11}{30} = 0$, i.e.

$$\lambda_{1,2} = -0.4333 \pm 0.4230j,$$

which are **identical** to the poles found in Task 2. The state-space model is therefore consistent with the transfer function.

6 Task 5 — Frequency Response (Bode)

Figure 2 shows the closed-loop frequency response $T(j\omega)$. The magnitude is essentially flat at the DC level ($20 \log_{10} 0.909 = -0.8$ dB) up to about 0.6 rad/s (the system bandwidth, set by ω_n), after which it rolls off; the phase falls smoothly from 0° towards -90° . Because the closed-loop poles lie in the left half-plane, the loop is stable with ample gain and phase margin.

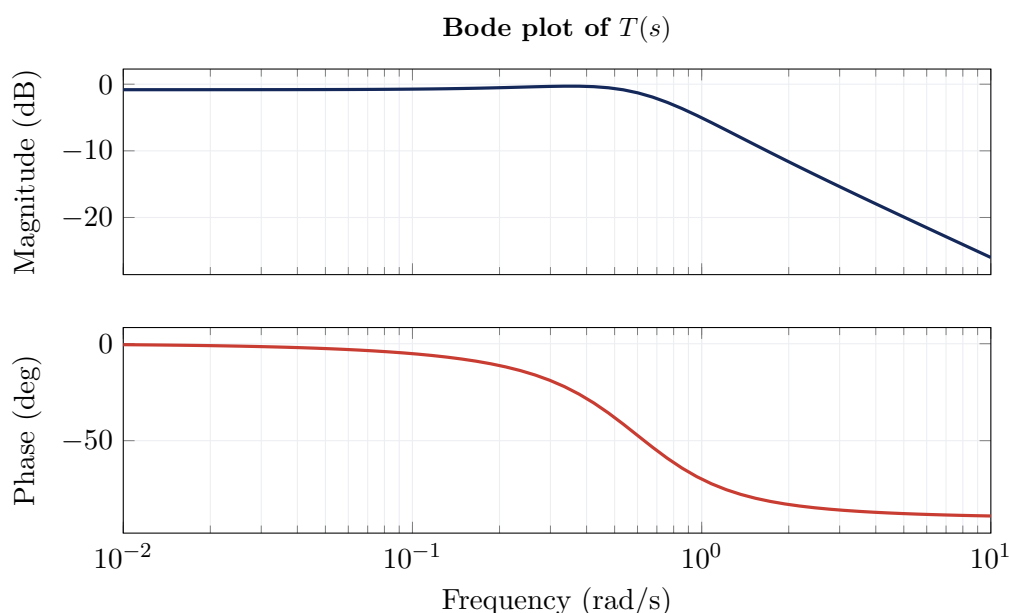


Figure 2. Closed-loop Bode magnitude and phase response.

7 Task 6 — Step Response

The unit-step response of $T(s)$ is shown in Figure 3. Its characteristics are summarised in Table 2. Because the DC gain is $T(0) = \frac{10}{11} = 0.909$, the output settles *below* the reference: the difference of 0.091 (9.1%) is the **steady-state error** that is inherent to a proportional speed-droop governor.

Table 2. Step-response characteristics of the closed-loop system.

Quantity	Value
Steady-state value (DC gain)	0.909
Steady-state error	0.091 (9.1%)
Peak value	0.988
Overshoot (relative to final)	$\approx 8.7\%$
Settling time ($\pm 2\%$)	≈ 9.2 s
Damping ratio ζ	0.716

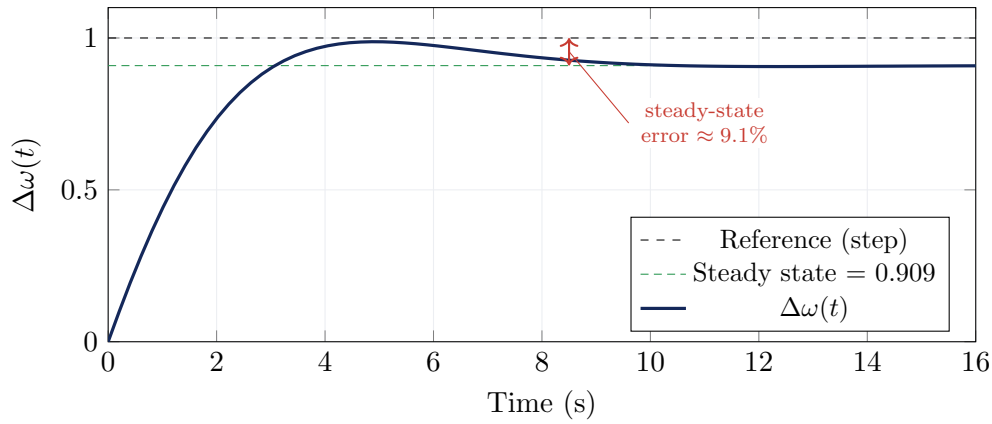


Figure 3. Closed-loop step response ($\pm$ a unit step in ω_{ref}).

Influence of reheat. The reheat block introduces the finite time constant T_{RH} and the zero at $s = -\frac{10}{15} = -0.667$. The zero produces the small initial overshoot, while T_{RH} slows the mechanical-power build-up; increasing F_{HP} (more prompt HP power) speeds the initial rise, as explored next.

8 Task 7 — Parameter Sensitivity (T_{RH})

Varying T_{RH} changes the *whole* closed-loop transfer function, not only its gain. Carrying the reheat constant symbolically gives

$$T(s) = \frac{3T_{RH}s + 10}{6T_{RH}s^2 + (6 + 4T_{RH})s + 11}.$$

Table 3 lists the dominant pole, damping ratio and settling time as T_{RH} varies from 0 to 10 s, and Figure 4 shows the corresponding step responses.

Table 3. Effect of the reheat time constant T_{RH} on the dominant dynamics.

T_{RH} (s)	Re(dominant pole)	Damping ζ	Settling T_s (s)
0	−1.833	— (real)	2.18
2.5	−0.533	0.623	7.50
5	−0.433	0.716	9.23
7.5	−0.400	0.809	10.00
10	−0.383	0.895	10.43

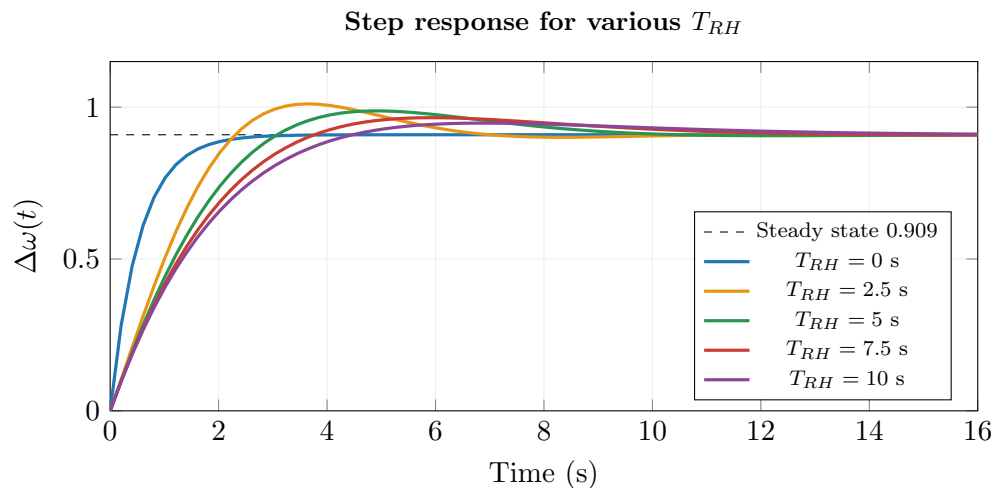


Figure 4. Step response as the reheat constant T_{RH} is varied. All curves converge to the same steady state (0.909); only the transient changes.

Comment. A larger reheat constant moves the dominant pole *towards* the imaginary axis (real part $-1.83 \rightarrow -0.38$), so the response becomes **slower** (settling time rises from about 2 s to 10 s). At the same time the damping ratio increases ($0.62 \rightarrow 0.90$), so the response becomes **less oscillatory** with smaller overshoot. Crucially — and unlike the original draft — the steady-state value is unchanged at 0.909 for every T_{RH} , because T_{RH} is a time constant and does not affect the DC gain $T(0) = 10/11$.

9 Task 8 — Python Verification

The following (corrected) Python performs the whole analysis: it derives and simplifies the transfer function symbolically, computes the poles/eigenvalues, and generates the Bode, step and sensitivity plots. The numerical results reproduce the hand calculations above exactly.

(a) Symbolic transfer function and poles

```
import sympy as sp
s = sp.symbols('s')
R, T_RH, F_HP, H, D = 0.1, 5, 0.3, 3, 1

G_c = (1/R) * (1 + F_HP*T_RH*s) / (1 + T_RH*s)  # governor + reheat turbine
G_p = 1 / (2*H*s + D)                          # generator-load swing
G_ol = G_c * G_p                                # open loop
T = sp.simplify(G_ol / (1 + G_ol))              # closed loop
print(sp.simplify(T))                          # -> (15*s + 10)/(30*s**2 + 26*s + 11)

poles = sp.solve(30*s**2 + 26*s + 11, s)
print(poles)                                   # -> -13/30 +- sqrt(161)*I/30 = -0.4333 +- 0.4230j
```

(b) State-space and eigenvalue check (corrected C matrix)

```
import numpy as np
A = np.array([[0, 1], [-11/30, -26/30]])
B = np.array([[0], [1]])
C = np.array([[10/30, 15/30]])  # corrected: reproduces the numerator 15s + 10
D = np.array([[0]])
print("eigenvalues:", np.linalg.eigvals(A))  # -0.4333 +- 0.4230j (== poles)
```

(c) Bode and base step response (corrected denominator)

```

from scipy import signal
num = [15, 10]
den = [30, 26, 11]           # corrected: constant is 11, not 10
sys = signal.TransferFunction(num, den)
w, mag, phase = signal.bode(sys)      # Bode data
t, y = signal.step(sys)              # step response; y -> 0.909 (DC gain)

```

(d) Parameter sensitivity (corrected transfer function)

```

import matplotlib.pyplot as plt
for T_RH in [0, 2.5, 5, 7.5, 10]:
    num = [3*T_RH, 10]           # numerator depends on T_RH
    den = [6*T_RH, 6 + 4*T_RH, 11] # full denominator depends on T_RH
    sysv = signal.TransferFunction(num, den)
    t, y = signal.step(sysv)
    plt.plot(t, y, label=f"T_RH = {T_RH} s")
plt.axhline(10/11, ls='--', color='k', label='steady state 0.909')
plt.xlabel("Time (s)"); plt.ylabel("dw(t)"); plt.legend(); plt.show()

```

10 Conclusion

The turbine–governor loop was analysed both by hand and in Python. The closed-loop transfer function is $T(s) = \frac{15s + 10}{30s^2 + 26s + 11}$, with a stable, under-damped complex-conjugate pole pair at $-0.4333 \pm 0.4230j$ ($\zeta = 0.716$, $\omega_n = 0.606$ rad/s). The state-space eigenvalues match these poles exactly once the output matrix is corrected to $C = [10/30 \ 15/30]$. The step response settles at 0.909, leaving the 9.1% steady-state error characteristic of proportional droop control, with about 8.7% overshoot and a 9.2 s settling time. Finally, increasing the reheat constant T_{RH} slows the response but improves damping, while leaving the steady-state gain unchanged — correcting the misleading behaviour of the original draft, in which the curves incorrectly settled at different values.